Mindlin sliding: friction 0.2

LAMMPS dem05-slip-mu02 | Tangential friction and spin | 28 October 2026

FAIL	0 of 4 runs pass both official 1% endpoint checks. vx passes: 0/4; spin passes: 4/4. This case is not validated against the official criteria.

Official endpoint check at 1.5 s

Run	dt (us)	vx (m/s)	Error (%)	omega_y (rad/s)	Error (%)	1% checks
CPU dt	10	0.704768783	1.332370	288.068070	0.823825	FAIL
GPU dt	10	0.704768783	1.332370	288.068070	0.823825	FAIL
CPU dt/2	5	0.704768781	1.332371	288.068072	0.823825	FAIL
GPU dt/2	5	0.704768781	1.332371	288.068072	0.823825	FAIL

Targets: vx = 5/7 = 0.714285714286 m/s; omega_y = vx/R = 285.714285714 rad/s. The official slip_cessation checker tests each relative error against 1%, at analytic_segment = 2 (step 150000).

Unchanged official inputs and mapping

Quantity	Official input / FunDEM configuration
Sphere / initial state	R = 0.0025 m; density 2700 kg/m^3; center z = 0.0024393 m; vx = 1 m/s; zero spin
Wall / gravity	Native floor z = 0; official upper z = 1 m is inactive and omitted; gravity = 1 m/s^2 downward
Contact materials	E = 10000 Pa; nu = 0.33 for both bodies; E* = 5611.04253170 Pa; G* = 1125.56841205 Pa
Law / damping	Hertz / Mindlin; sliding friction 0.2; restitution 1; automatic damping 0; linear-stiffness and manual damping overrides off
Other terms	Rolling/torsional friction and global damping = 0; no fitted coefficients or solver changes
Time and output	1.5 s; dt = 10 / 5 us; 150,000 / 300,000 steps; 1501 native frames per run at 1 ms; CSV every step

Ideal solution and contact convention

I = (2/5) m R^2; vx = 1 - 0.2 t; omega_y = 200 t.
s_R = vx - R omega_y = 1 - 0.7 t; t* = 2/(7 mu g) = 1.428571428571 s.
Friction 0.2 doubles t* versus 0.4 but preserves ideal final vx = 5/7 and omega_y = (5/7)/R. The official criterion tests the endpoint velocities.

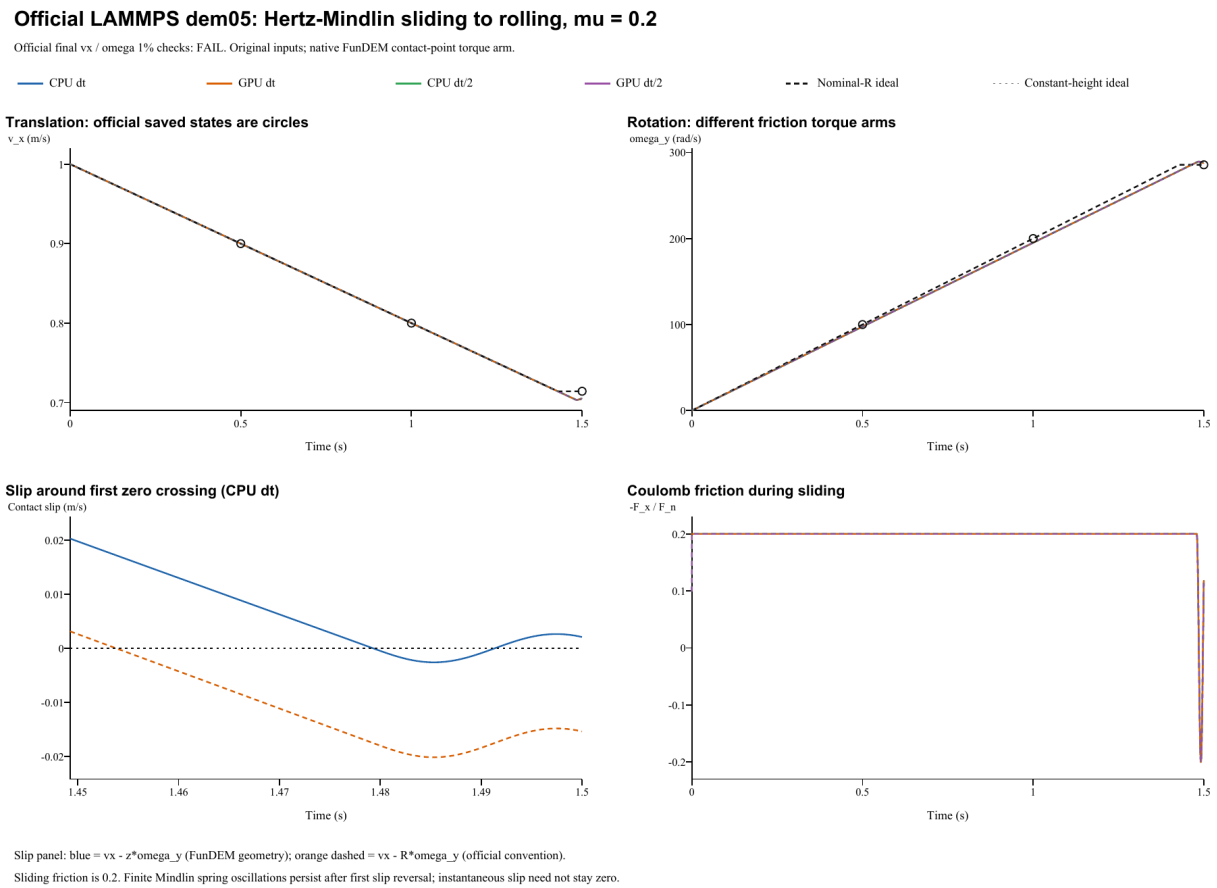
FunDEM uses a = R - delta = z for slip and torque; LAMMPS uses nominal R. Initial overlap is 60.7 um (2.4280% of R). At Hertz equilibrium h = 0.0024393431 m, the supplemental constant-arm model predicts vx = 0.70415584 m/s, omega_y = 288.666170 rad/s and t = 1.47922082 s. This is a diagnostic, not an adjusted target. A finite Mindlin spring oscillation remains: the endpoint values in the table differ from this approximation, and sustained exact pure rolling at 1.5 s is not established.

Reference provenance and boundary mapping

Official YAML and saved states | Analytic checker. Commit f9720ce2d176d615495d1d6c31349b1f71be1c73; YAML blob ccecc980cfbec0aecff7827c313801c7c06d07a0. LAMMPS was not rerun locally; epsilon 1e-10 is not a cross-code tolerance. The isolated sphere on a uniform plane uses unwrapped native x/y; x is wrapped to [-0.05, 0.5] m for saved-state comparison.

Curves and retained evidence

Real native trajectories, ideal targets and pinned official saved states



Numerical and contact checks

Maximum CPU/GPU history differences: 1.37e-14 m/s in velocity and 5.74e-12 rad/s in spin. Halving dt changes the CPU endpoint vx by 1.44e-09 m/s and omega_y by 1.4e-06 rad/s. Relative to the retained friction-0.4 Mindlin case, the first geometric-slip zero occurs 2.00002643 to 2.00002646 times later; the ideal inverse-friction transition-time factor is 2. This diagnostic does not change the official endpoint targets.

Mean sliding abs(Ft)/Fn = 0.20000000 to 0.20000000; maximum torque-derived arm error = 4.34e-19 m. A first slip zero can precede tangential spring oscillations and does not prove sustained exact pure rolling. Backend agreement and dt convergence do not override the official outcome.

Complete configured native results are retained

CPU dt: run.json | GPU dt: run.json | CPU dt/2: run.json | GPU dt/2: run.json

1501 native frames per run, 6004 total, including run.json, project.json, configured frames/contacts, input projects, audits and backend logs. Output every 1 ms (100 base steps / 200 half steps). CSV separately records all steps: 900004 trajectory rows and 900000 contact rows. Native frames were not saved at every integration step.

Run manifest | Parameter mapping | Analysis JSON | Checkpoint CSV | Contact audit | Complete result ZIP

Execution and validation scope

Real FunDEM SimulationSession with linked 0.4.3 CPU/CUDA native libraries; no solver changes or fitted inputs. Byte identity with the installed GUI executable was not established. Earlier dem05 outcomes remain unchanged.

This case tests soft Hertz/Mindlin friction, spin-up and approach to rolling. Viscous damping is zero: a limiter acting on nonzero tangential damping is not tested. Nonzero rolling resistance, torsion, damping and many-particle flow require other cases.